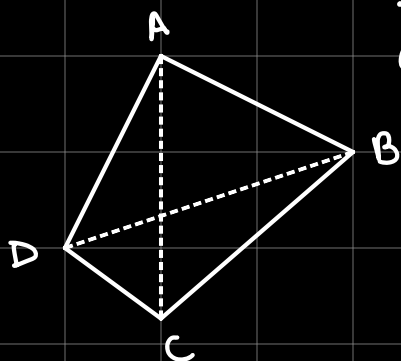


FUNDAMENTOS

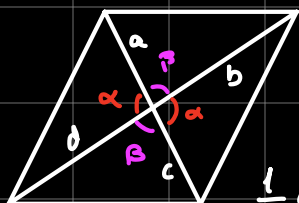
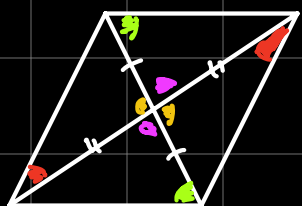


I. Se as diagonais $\overline{AC}$ e $\overline{BD}$ tem o mesmo comprimento e são $\perp$ uma à outra, ABCD é losango

X

II. Se as diagonais dividem o quadrilátero em 4 Triângulos de Área igual. Verdade

(Diagonais se cruzam ao meio são Paralelogramos)



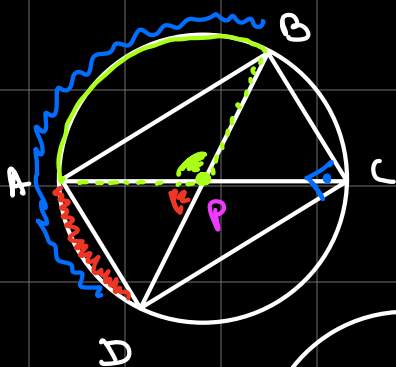
$$\alpha + \beta = 180^\circ$$

$$\sin \alpha = \sin \beta$$

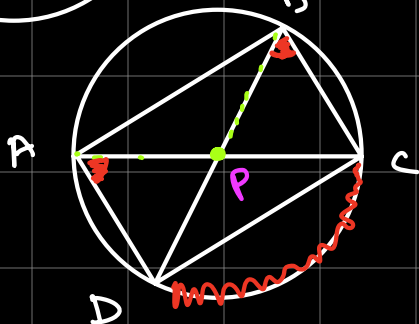
$$\frac{1}{2} a \cdot d \sin \alpha = \frac{1}{2} a \cdot b \sin \beta$$

$$\text{então } d = b$$

$$\text{e assim } a = c$$



III) Se o ponto de Intersecção das diagonais AC, BD é o Centro do Circulo qm circunscreve ABCD então ABCD é retângulo Verdade



$$\hat{A} + \hat{C} = 180^\circ$$

$$\hat{B} + \hat{D} = 180^\circ$$

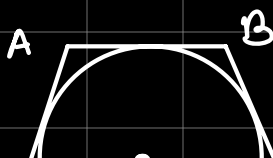
TEOREMA DAS Cordas

$$AP \cdot PC = BP \cdot PD$$

Teorema de Ptolomeu

$$AC \cdot BD = AD \cdot BC + AB \cdot CD$$

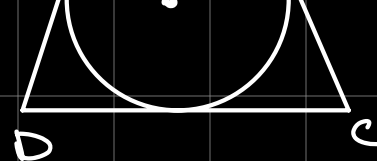
Teorema de Pitag



$$AB + CD = AD + BC$$

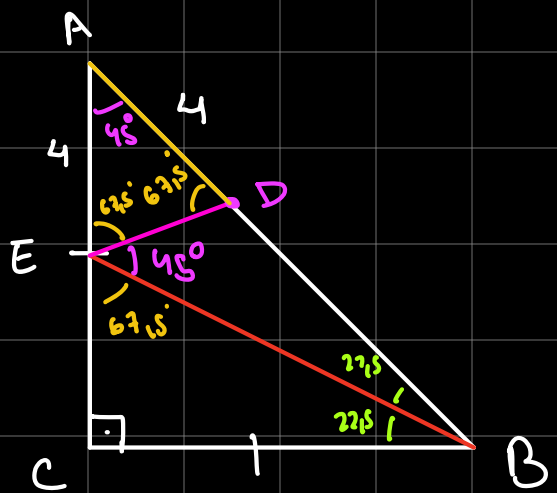


Se um trapézio é circunscritível ele é isóceles



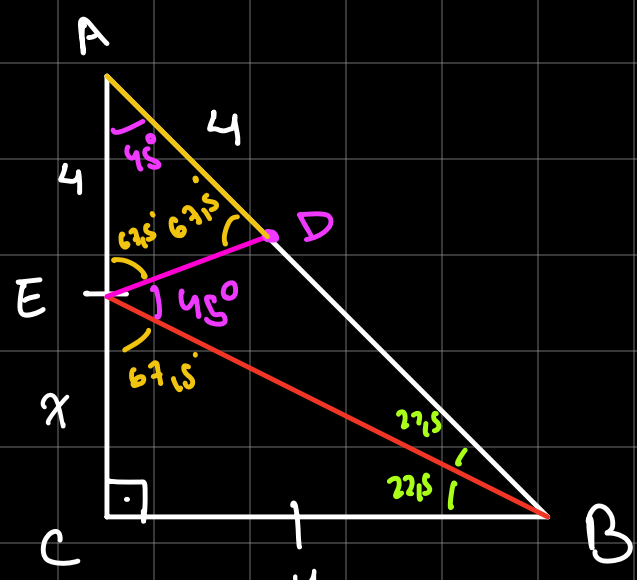
é polígono Circunscrito

Triângulo Retângulo

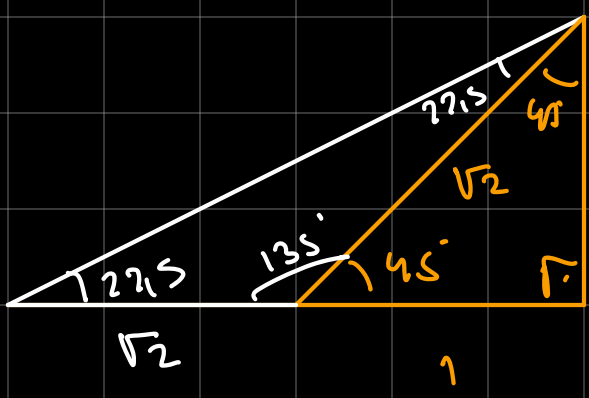


$$x + 4 = y$$

$$\frac{x}{y} = \sqrt{2} - 1$$



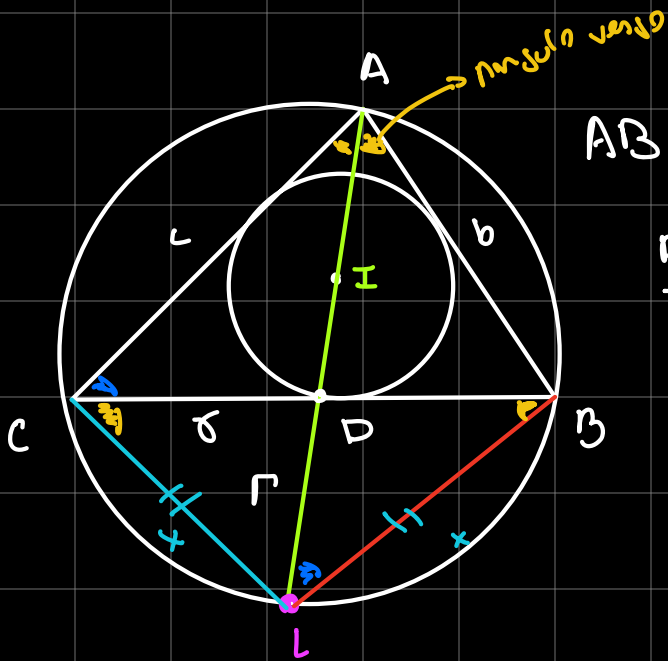
$$2y^2 = (BD + 4)^2$$



→ Divisor de Ângulos

⚠ lembrar do Divisor de ângulos

RAZÕES E PROPORÇÕES



$$AB + AC = 2BC$$

$$\frac{BL}{AL} = ?$$

$$\frac{BL}{r} = \frac{AC}{\delta} = \frac{1}{2} \quad \checkmark$$

$$TBI(\Delta ABC)$$

$$\frac{AB}{AC} = \frac{BD}{DC} \Rightarrow BD = \alpha_c$$

$$BD + DC = BC$$

$$\alpha(c \cdot b) = \beta c$$

$$x = \frac{BC}{BD + DC}$$

$$\frac{CL}{AL} = \frac{BD}{AB} \Rightarrow CL = BL$$

$$B D = \alpha_c = \frac{c}{2}$$

$$\frac{B_L}{A_L} = \frac{C_{12}}{C} = \frac{1}{2}$$

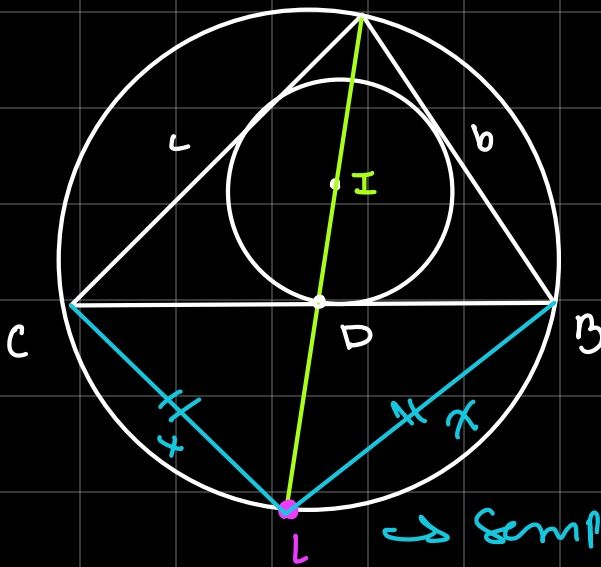
Per PTO/amen

$$xb + cx = AL \cdot a$$

$$\chi(b+c) = A^L \cdot a$$

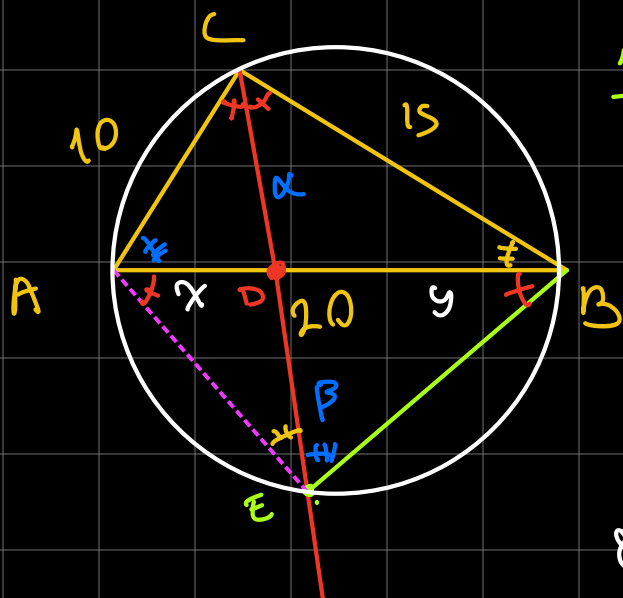
$$\frac{x}{AL} = \frac{a}{b+c} = \frac{1}{2}$$

é importante lembrar que



→ sempre
é isóceles

17A 2017



$$\frac{10}{15} = \frac{x}{y}$$

$$x + y = 20$$

$$\frac{2}{3} = \frac{x}{y}$$

$$2x + \frac{3}{2}x = 20$$

$$2y = 3x$$

$$\frac{8x}{2} = 20 \quad \text{?}$$

$$x = 8$$

$$y = 12$$

$$8 \cdot 12 = \alpha \cdot \beta$$

$$\alpha^2 = bc - mn$$

$$\alpha^2 = 10 \cdot 15 - 8 \cdot 12 = 54$$

$$\alpha = 3\sqrt{6}$$

$$\beta = \frac{32}{\sqrt{6}}$$

$$\alpha + \beta = \frac{32}{\sqrt{6}} + 3\sqrt{6}$$

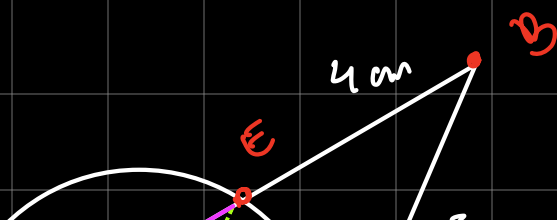
$$\alpha + \beta = \frac{16\sqrt{6}}{3} + 3\sqrt{6} = \frac{25\sqrt{6}}{3}$$

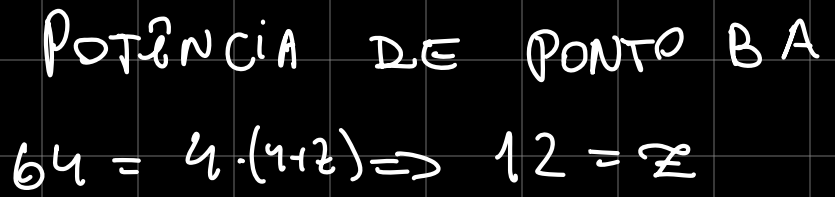
Importance

→ lembrar Stewart → $x^2 = bc - mn$ (BTZ INTERNA)

→ lembrar BTZ Interna → $b/c = m/n$

DS)

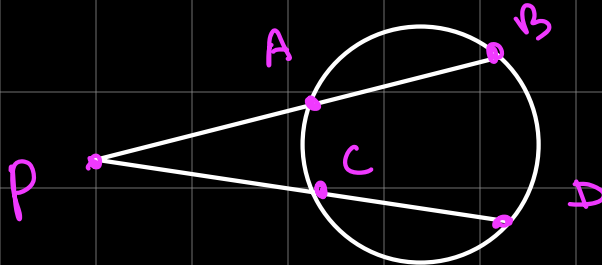



$$36 = 4 \cdot (4+1) \Rightarrow y = 5$$
$$eAC = 9$$
$$\cos \theta = \left(\frac{14^2 - 9^2 - 16^2}{2 \cdot 9 \cdot 16} \right) = \frac{47}{96}$$

$$DE^2 = y^2 + z^2 - 2yz \cos \theta$$

Lembrar potencia de PONTO

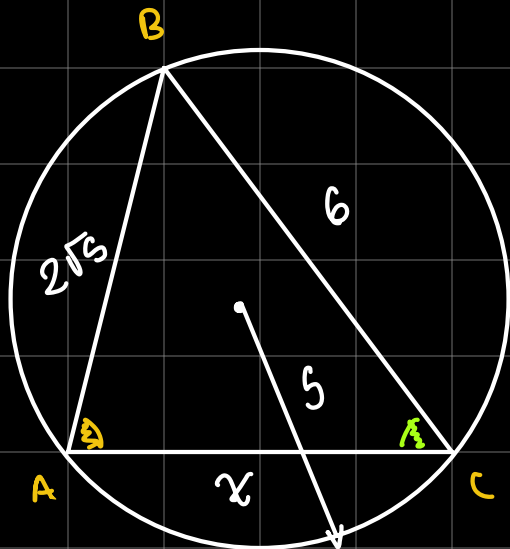
$$p_A \cdot p_B = p_T^2$$



Leis dos Senos e Cossenos

$$\frac{2\sqrt{5}}{\sin \hat{C}} = \frac{6}{\sin \hat{A}} = 10$$

$$\sin \hat{C} = \frac{\sqrt{5}}{5} \quad \cos \hat{C} = \frac{2\sqrt{5}}{5}$$



Lei dos Cossenos no $\triangle ABC$

$$x^2 + 6^2 - 2 \cdot x \cdot 6 \cos \hat{C} = (2\sqrt{5})^2$$

$$x^2 + 36 - 12x \cos \hat{C} = 20$$

$$x^2 + 36 - \frac{24\sqrt{5}}{5}x = 20$$

$$5x^2 - 24\sqrt{5}x + 80 = 0$$

$$\Delta = 24^2 \cdot 5 - 4 \cdot 5 \cdot 80$$

$$\Delta = 1280$$

$$\sqrt{\Delta} = 16\sqrt{5}$$

$$x = \frac{24\sqrt{5} \pm 16\sqrt{5}}{10} \Rightarrow \begin{cases} 4\sqrt{5} \\ \frac{4\sqrt{5}}{5} \end{cases}$$

$$4\sqrt{5} > 6$$

ou seja é

o maior lado



Tem que testar

como $\frac{4\sqrt{5}}{5} < 6$ para este valor $\hat{B}$ é agudo

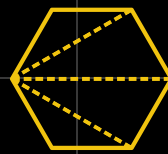
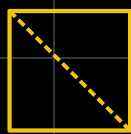
$$\text{já para } 4\sqrt{5} \rightarrow (4\sqrt{5})^2 = (2\sqrt{5})^2 + 6^2 - 2 \cdot 6 \cdot 2\sqrt{5} \cdot \cos \hat{B}$$

$$16 \cdot 5 = 20 + 36 - 24\sqrt{5} \cos \hat{B}$$

$$80 - 36 = -24\sqrt{5} \cos \hat{B}$$

$$-\frac{24}{24\sqrt{5}} \Rightarrow \cos \hat{B} = -\frac{\sqrt{5}}{5} \rightarrow \text{obtusos}$$

Polígonos



Considere as afirmações

$$\sum \alpha_i = (n-2) \cdot 180^\circ$$

I) Se P é polígono convexo de n lados iguais, então P é regular

II) Seja P um polígono convexo de 6 lados. Se seus ângulos internos, listados em ordem crescente formam um P.A. então a soma do menor e do maior ângulo interno é 240°

III) $\exists$ polígono convexo de 100 lados cujos ângulos internos listados em ordem crescente formam PA de razão 1

Pontos Notáveis

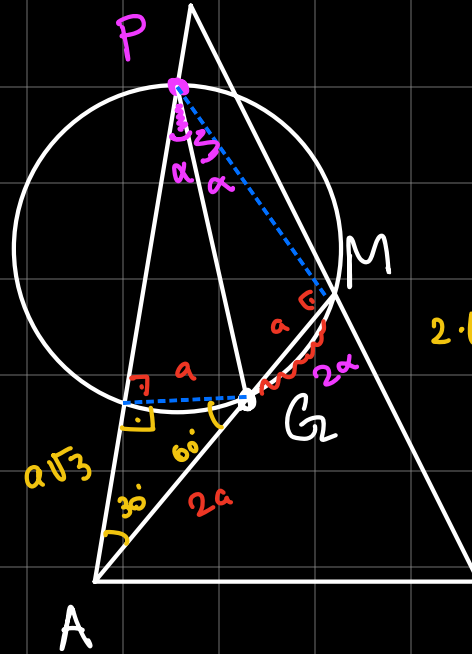
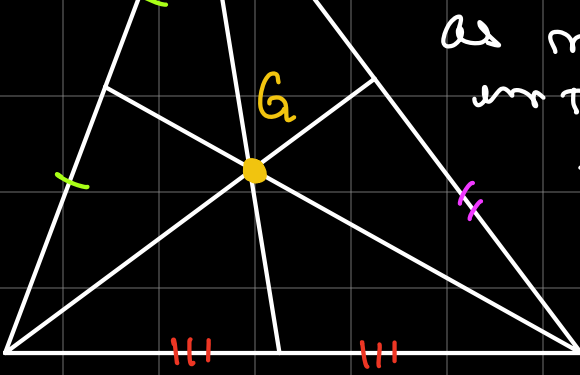
• Mediana Baricentro



Centro de Massa

Divide a mediana na razão de $2:1$

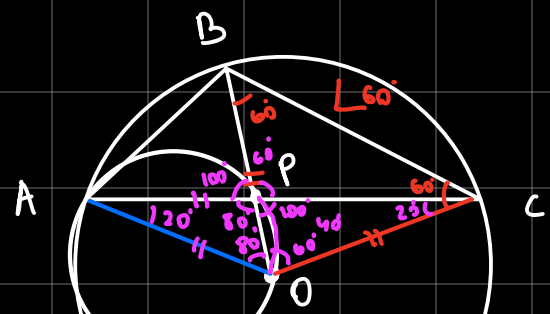
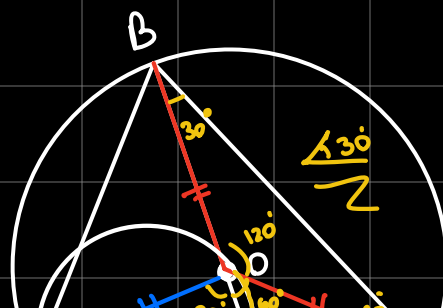
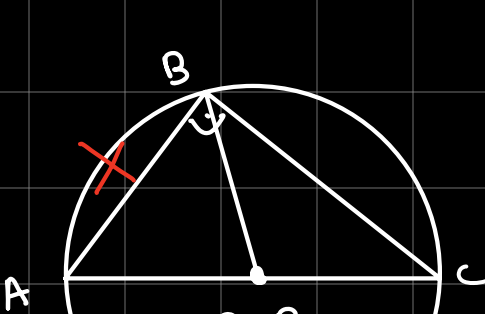
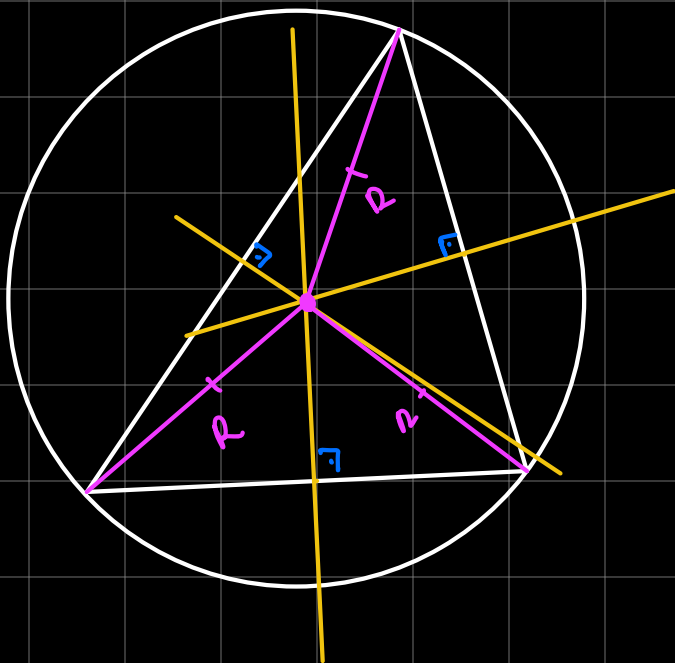
As medianas separam o triângulo em 6 triângulos de mesma área



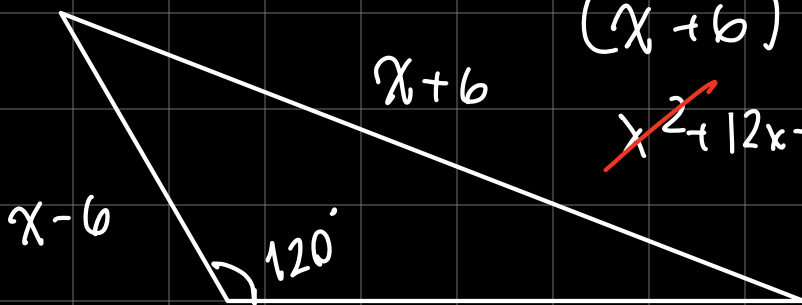
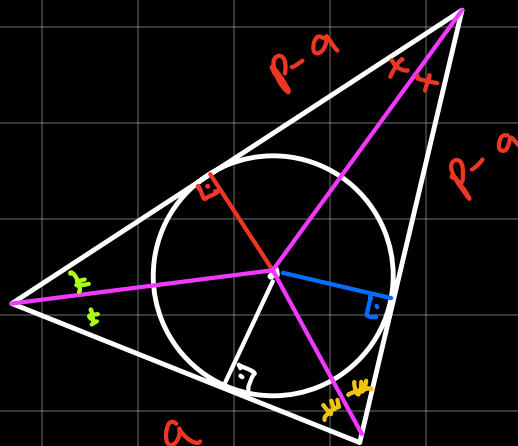
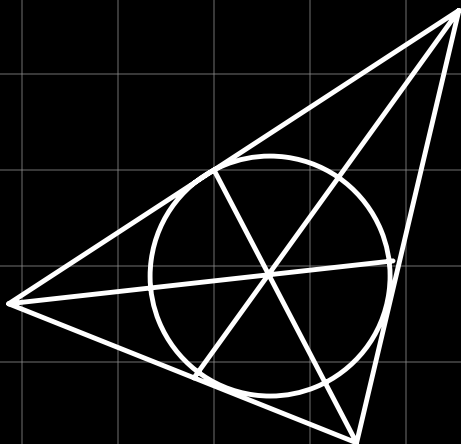
$$\begin{aligned} 2 \cdot (90^\circ - \alpha) + 60^\circ &= 180^\circ \\ 180^\circ - 2\alpha + 60^\circ &= 180^\circ \\ 2\alpha &= 60^\circ \\ \alpha &= 30^\circ \end{aligned}$$

• Circumcentro (Mediotrizes)

$$R = \frac{abc}{4[\Delta]}$$



① Incontro



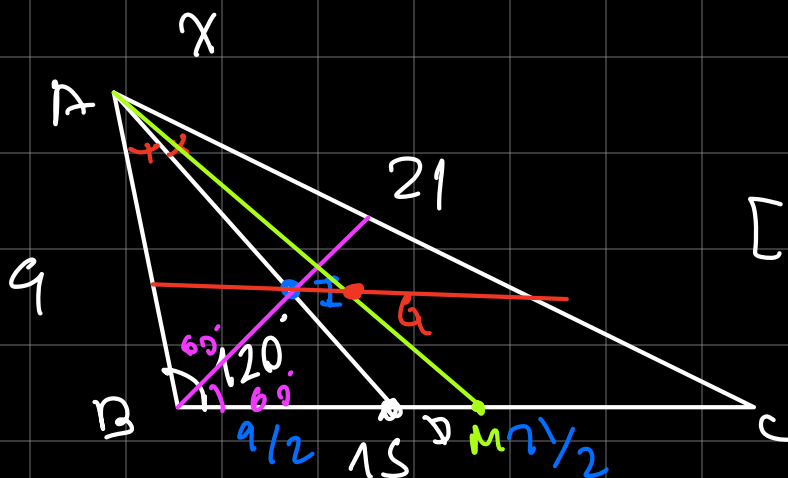
$$(x+6)^2 = (x-6)^2 + x^2 - 2x(x-6)\cos 120^\circ$$

$$\cancel{x^2} + 12x + 36 = \cancel{x^2} - 12x + 36 + x^2 + x^2 - 6x$$

$$2x^2 - 30x = 0$$

$$2x(x-15) = 0$$

$$x = 15$$



$$[ABC] = \frac{1}{2} \cdot 9 \cdot 15 \sin 120^\circ$$

$$[ABC] = 135 \frac{\sqrt{3}}{4}$$

$$TB I = \frac{BD}{DC} = \frac{9}{21} = \frac{3}{7}$$

$$TB I = \frac{AI}{ID} = \frac{9}{9/2} = 2 \quad AI = 2 ID$$

Se a proporção $AI : ID$ correja a
mesma proporção $1 : 2$

de $AG : GD \rightarrow IG \parallel BC$ pela recíproca
do Teorema de Tales

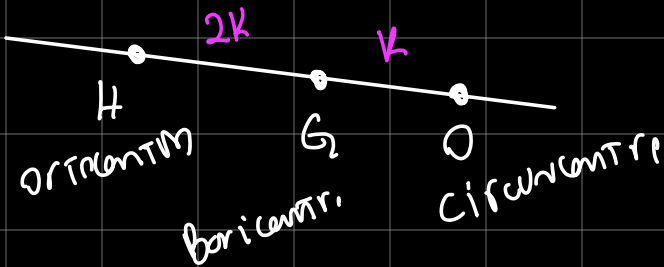
$$DM = BM - DB = \frac{15}{2} - \frac{9}{2} = 3$$

$$\triangle AIG \sim \triangle ADM$$

$$\frac{IG}{DM} = \frac{AG}{AM} \Rightarrow \frac{IG}{3} = \frac{2}{3}$$

$$\underline{IG = 2}$$

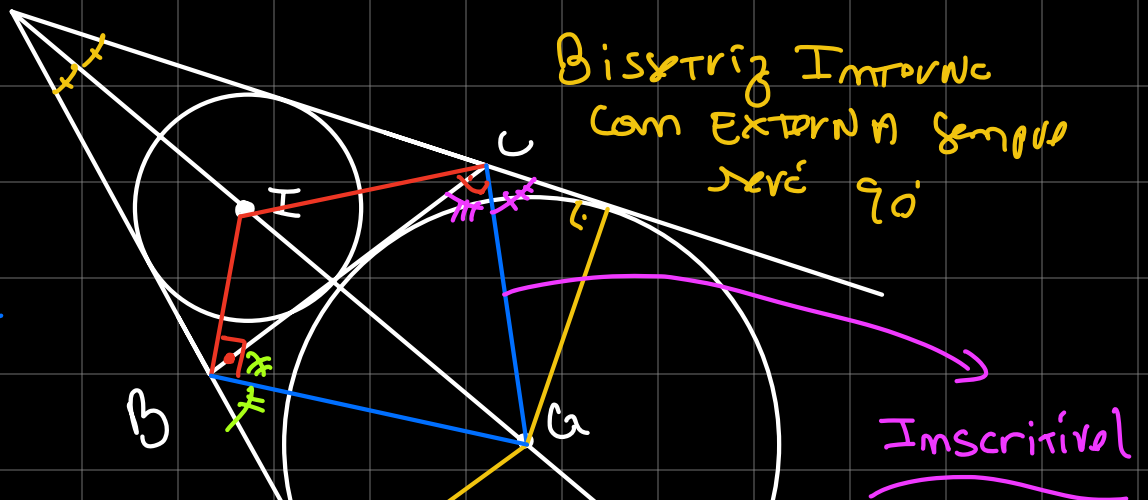
Nota De euler



EXCENTRO (EXIMCENTRO)

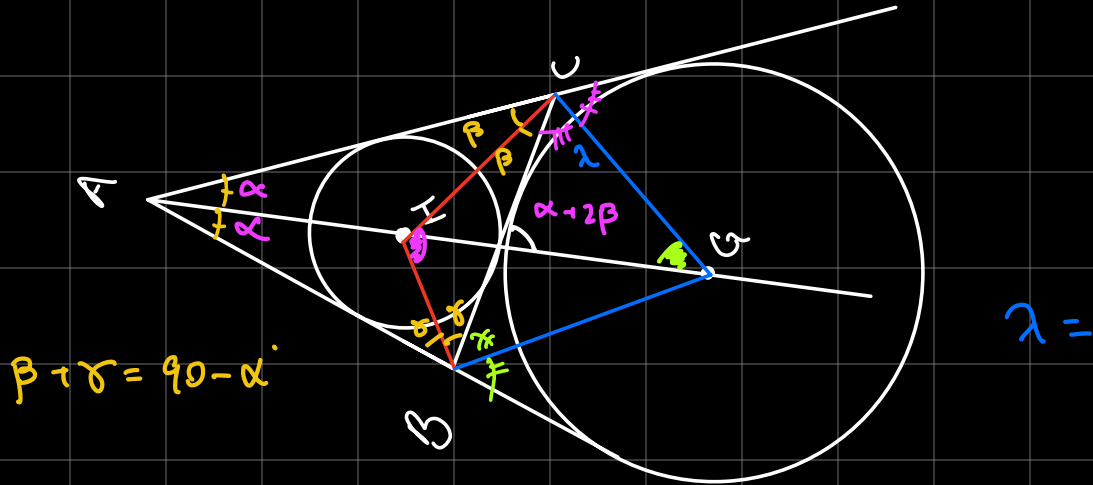
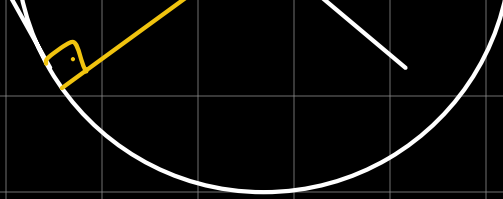
é o encontro de duas bts externas e 1 interna

A



$$r_{ex} = \frac{p r_{ins}}{p - a}$$

$$\angle BIC + \angle AGC = 180^\circ$$



$$\beta + \gamma = 90 - \alpha$$

$$180 - \beta - \gamma + (180 - \alpha - 2\beta - \lambda) = 180$$

$$90 + \alpha +$$

$$\lambda =$$